1. Holomorphic maps on $\mathbb{C}^n$

Recall the following notations.

- $\phi : \mathbb{C}^n \to \mathbb{R}^{2n}$ the canonical homeomorphism taking $(z_1, \dots, z_n)$ onto $(x_1, \dots, x_n, y_1, \dots, y_n)$, where $z_j = x_j + iy_j$ for $j = 1, \dots, n$.
- J_0 is the canonical almost complex structure on $\mathbb{R}^{2n}$.

Definition 1.1. A map $f : U \to \mathbb{R}$ defined on an open subset U of $\mathbb{C}^n$ is *smooth* if $f \circ \phi^{-1}$ is a smooth map on $\phi(U)$. $f : U \to \mathbb{C}$ is *smooth* if its real and imaginary parts are smooth.

Definition 1.2. A smooth map $f : U \to \mathbb{C}$ defined on an open subset U of $\mathbb{C}^n$ is *holomorphic* if $\tilde{f} = f \circ \phi^{-1}$ is J_0 holomorphic, that is, $d\tilde{f} \circ J_0 = i\, df$, or equivalently $d\tilde{f} \circ J_0 = J_0 \circ d\tilde{f}$, where J_0 on the right hand side is the canonical almost complex structure on $\mathbb{R}^2$.

A smooth map $f : U \to \mathbb{C}^m$ is *holomorphic* if each component function $f_j : U \to \mathbb{C}$ of f is holomorphic, $j = 1, \dots, m$.

2. Complex manifolds and associated almost complex structure

Let M be a complex manifold of dimension n. Then there is an open covering $\{U_\alpha\}$ of M together with homeomorphism $\varphi_\alpha : U_\alpha \to V_\alpha$, where V_α is open in $\mathbb{C}^n$, such that

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

is a biholomorphism. We shall denote $\varphi_\beta \circ \varphi_\alpha^{-1}$ by $g_{\alpha\beta}$.

(a) Let $\tilde{\varphi}_\alpha = \phi \circ \varphi_\alpha$ for all α. Prove that $\{U_\alpha, \tilde{\varphi}_\alpha\}$ is an atlas for M. In other words, it defines a smooth manifold structure on M.

(b) Show that the transition functions $\tilde{g}_{\alpha\beta} : \tilde{\varphi}_\alpha(U_\alpha \cap U_\beta) \to \tilde{\varphi}_\alpha(U_\alpha \cap U_\beta)$ satisfy the equation $d\tilde{g}_{\alpha,\beta} \circ J_0 = J_0 \circ d\tilde{g}_{\alpha\beta}$.

(c) Since M is now a smooth manifold (of dimension $2n$) we can talk about its tangent bundle TM, which has an atlas $\{TU_\alpha, d\tilde{\varphi}_\alpha\}$. Show that there is a unique almost complex structure J on TM defined by $J_\alpha(v) = (d\tilde{\varphi}_\alpha)^{-1} J_0\, (d\tilde{\varphi}_\alpha)(v)$ for all $v \in TU_\alpha$.

Definition 2.1. A smooth map $f : M \to N$ between two complex manifolds is *holomorphic* if $\psi f \varphi^{-1}$ is holomorphic for all charts (U, φ) of M and (W, ψ) on N satisfying $f(U) \subset W$.

(d) If (U, φ) is a chart on a complex manifold M then prove that φ is holomorphic. Furthermore, if $\varphi = (z_1, \dots, z_n)$, then the coordinate functions $z_j = p_j \circ \varphi$ on U are also holomorphic.

(e) Show that, if $f : M \to N$ is holomorphic then $df \circ J_M = J_N \circ df$, where J_M and J_N are respectively the canonical almost complex structure on M and N.